

OPEN MERCATO X KRONOS X AZURE

Your storefront already streams the future. Forecast it. Act on it.

Every order in Open Mercato is a live event. Turn that stream into **real-time demand forecasts** with Kronos on Azure — and get an **automatic alarm** the moment the model stops matching reality.

`order.*` events

GMV / minute

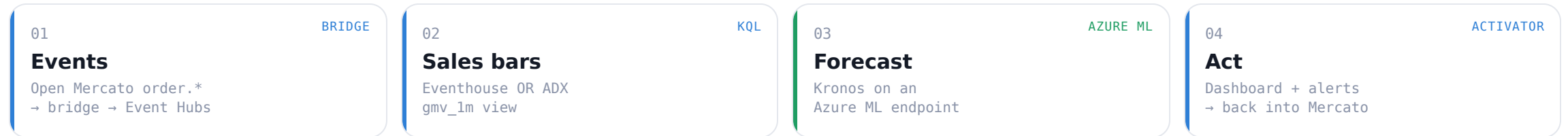
Fabric RTI or Azure-only

THE ARCHITECTURE

One event bridge. Then the same real-time loop.

Open Mercato publishes domain events; a small **bridge** forwards them to Azure. From there it's the markets pattern, retargeted to demand.

■ FABRIC / AZURE DATA PLANE · ■ AZURE ML MODEL PLANE



Option A · Microsoft Fabric RTI

Eventstream → Eventhouse → Real-Time Dashboard → Data Activator. No-code, governed, one platform.

Option B · Azure-only

Event Hubs → Azure Data Explorer → Azure Functions → Monitor. **Same KQL scripts** — ADX and Eventhouse both speak it.

Forecast demand — and know the instant it breaks.

DEMAND YOU CAN TRUST

~70.0%

Directional accuracy

30-min GMV horizon, in-regime, at ~2.76% median error. Surges become **signals** — pre-stock, pre-scale.

HONEST ABOUT LIMITS

0

Outages predicted

No model foresees a checkout outage or a viral spike. The forecast keeps projecting normal demand — as it should.

THE GUARDRAIL

~5 min

To auto-detect the break

Forecast-vs-reality error breaches its learned band and fires automatically — a stockout, outage or fraud wave, flagged fast.

LIVE DEMO

One store-day.

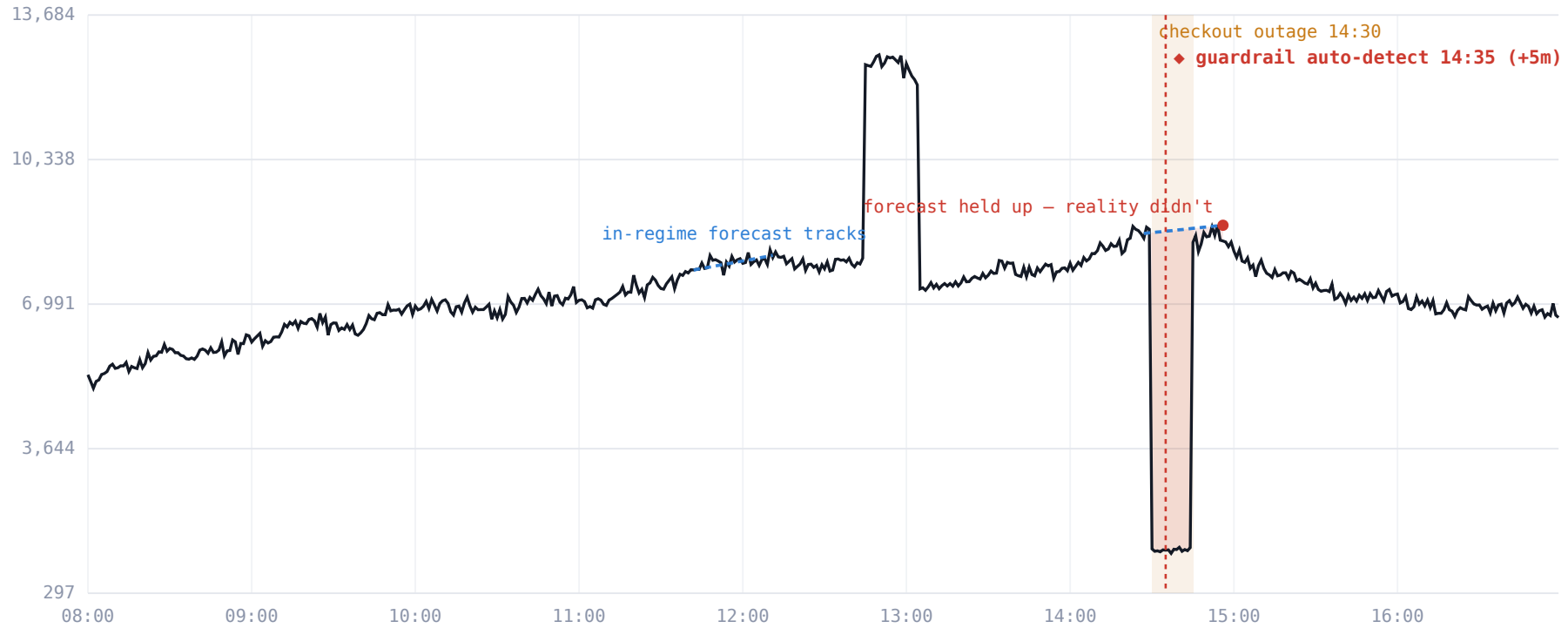
A synthetic Open Mercato tenant **store_hydra**: **66,401 order events** → **540 sales bars**, a Kronos forecast every minute (140 runs, 30-min horizon), a **flash sale**, and a **checkout outage**. The next slides are the real output.

Fixed-seed synthetic data — illustrative of the loop, not a live-store performance claim.

GMV / minute — actual vs forecast

Kronos tracks demand in-regime; the forecast issued just before the **14:30** checkout outage (red divergence) is exactly what the guardrail is built to catch.

— actual GMV - - forecast ■ error



70.0%

DIRECTIONAL ACCURACY

2.76%

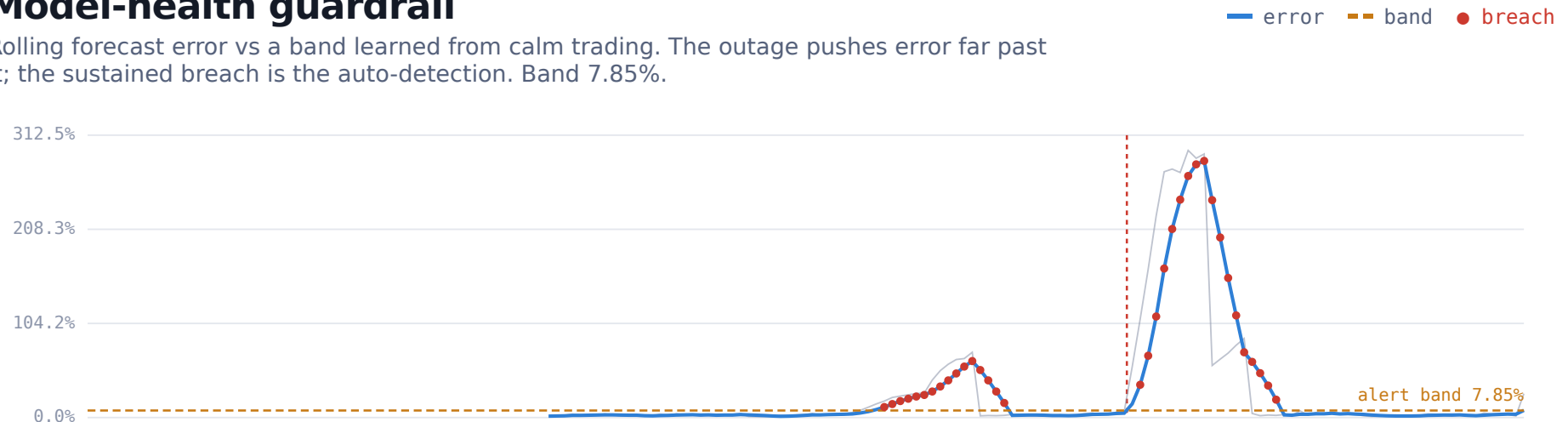
MEDIAN ERROR (MAPE)

61.1%

SIGNAL HIT-RATE · 95

Model-health guardrail

Rolling forecast error vs a band learned from calm trading. The outage pushes error far past it; the sustained breach is the auto-detection. Band 7.85%.



+5 min

TO AUTO-DETECT THE OUTAGE

14:30

CHECKOUT OUTAGE

ACT ON IT

A flash sale becomes a signal.

Data Activator raises a demand signal when the forecast expects a large move — a surge to pre-stock and pre-scale for, a drop to investigate. Sample of the flash-sale surge (of 95 signals @ 61.1% hit).

TIME	CALL	EXPECTED	REALIZED	RESULT
12:59	UP	+15.0%	-37.8%	· miss
13:02	UP	+15.7%	-37.2%	· miss
14:23	UP	+2.7%	-2.4%	· miss
14:26	UP	+2.3%	-2.0%	· miss
14:29	UP	+2.8%	-3.4%	· miss
15:05	UP	+2.3%	-9.7%	· miss
16:29	UP	+52.0%	-4.1%	· miss

IN ONE LINE

Orders → forecast → act. One loop, on Fabric or Azure.

Replays into Fabric Eventhouse or Azure Data Explorer: `replay_all.kql`

Kronos runs on commerce **sales bars** (fine-tune for accuracy); the durable value is the real-time loop + guardrail. Not a Microsoft or Open Mercato product.